

Review retrospective:
the J-function asymptotic-equivalence wrapper — closing the arc

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Abstract

A soundness review of `Laplace/OneD/JFunctionAsymptoticEquivalence.lean`: the user-facing asymptotic-equivalence form $J_k(t) \sim \pi(0) \cdot Z_k(t)$, built from an unconditional ratio limit $J_k/Z_k \rightarrow \pi(0)$ and the `IsEquivalent` packaging. Result: **a clean fidelity pass with two small dead-code removals** — a dead `open Real` and a linter-confirmed dead local. This is the last file of the J-function arc.

1 Setting

This wrapper turns the generic- k J-function asymptotic (from `JFunctionMonomial`) into the form the grammar paper uses: an asymptotic equivalence $J_k \sim \pi(0)Z_k$. It is the fourth and final file of the J-function arc (quartic, sextic, generic, and this equivalence wrapper).

2 What was reviewed

Two public theorems: `kth_jfunction_div_partition` (the unconditional ratio limit $J_k(t)/Z_k(t) \rightarrow \pi(0)$, the building block) and `kth_jfunction_isEquivalent` (the $\sim$ [atTop] form, valid for $\pi(0) \neq 0$, via `isEquivalent_iff_tendsto_one`).

3 Fidelity / soundness findings

Sound; no mismatch. The ratio limit is correct: writing $\alpha = 1/(2k)$ and $C = (1/k)((2k)!)^\alpha \Gamma(\alpha)$, the proof uses $t^\alpha J_k \rightarrow \pi(0)C$ (the N1 asymptotic) and $t^\alpha Z_k = C$ (the partition closed form) to get $J_k/Z_k = (t^\alpha J_k)/C \rightarrow (\pi(0)C)/C = \pi(0)$, with C shown positive before cancellation. The unconditional claim genuinely is unconditional (no $\pi(0) \neq 0$ is smuggled in — the ratio tends to 0 when $\pi(0) = 0$). The `IsEquivalent` theorem has the right orientation and comparator, supplies the eventual non-vanishing of $\pi(0)Z_k$ (from $Z_k > 0$ via the positive closed form, plus $\pi(0) \neq 0$), and reduces correctly to the ratio-to-one statement.

One report-only note: both theorems take a global `Continuous prior` hypothesis, then derive `.aemeasurable/.continuousAt` to call the minimal-hypothesis generic — the same non-minimal hypothesis carried by `JFunctionQuartic`. Weakening it to `AEMeasurable + ContinuousAt 0` is a forward tide (signature change), not a review edit.

4 Simplifications applied

Two dead-code removals. A dead open Real (every `Real.exp/Real.Gamma/Real.rpow_*/Real.div_rpow` is qualified; `MeasureTheory/Filter/Asymptotics/Topology` carry the load), and a dead local hypothesis `h $\alpha$ _ne` : $\alpha \neq 0$ in the ratio theorem (no denominator is α ; the `field_simps` use $t^\alpha \neq 0$ and $C \neq 0$). The local is a derived `have`, not a public binder, so its removal carries no unused-public-hypothesis cascade (the hazard from the Harmonic review); the `unusedVariables` linter confirmed it dead. Rebuild green, both public signatures byte-identical, GPT-affirmed.

5 Disagreements and open questions

None.

6 Lessons

- **Dead local `have` vs dead public binder — different risk classes.** Removing the derived local `h $\alpha$ _ne` is safe in a way that removing a dead *proof block* which consumes a public hypothesis (the Harmonic case) is not: a local `have` has no signature consequence, so the only failure mode is an implicit tactic dependence, caught by the rebuild. The `unusedVariables` linter's silence after removal is the clean confirmation.
- **The J-function arc carried a consistent non-minimal-hypothesis seam.** Three of the four files (quartic, this wrapper) take global `Continuous`, while the generic and the sextic use the minimal class. Reviewing the arc end-to-end made the seam legible as a single forward-tide cleanup rather than four unrelated notes.

7 Follow-ups

- **Forward tide (arc-wide):** migrate `JFunctionQuartic` and this wrapper to the minimal hypothesis class `AEMeasurable + ContinuousAt 0` that the generic already supports.
- The J-function arc is now fully reviewed. Remaining laplace surface: the `TwoD/` families (`KthKth*`, `QuarticSextic`, `AddSeparable`, `SemiDegenerate`), assorted `OneD/` files (`NearDegenerate`, `UniversalAsymptotics`, `TailBound`, ...), and the large `Multi/Covariance*` trio (sectioned approach).